

OCR/LaTeX: blackman_ch04_dynamic_targets_start

: Blackman_Popoli_Design_and_Analysis_of_Modern_Tracking_Systems

PDF: 229–237

: ; ; (white-noise);

: OCR. , PDF.

4.1 , . . , , — . , - , . , , . 4.4 , . 199 , , . 4.2 4.3 , . , . 4.4 , . 199

200 Design and Analysis of Modern Tracking Systems IMM () . 4.5 IMM. 4.6 ,
 ' IMM. . 3D- 4.7. , ' . (TBM), SCUD, IMM-. , 4.8
 IMM TBM-. 4.9 . 4.2 Q . Q (-), . Q , [1]. , x , v ,
 , , a . (7). , , [2, 3]. , , , . , [3] [4], , (). . , , '
 . , EKF. 4.2.1 [2] , ,

‘Modeling and Tracking Dynamic Targets 201 $a(k+1) = pma(k) + /1- p2, omr(h)$ (4.1)
 $Pm=e Pl, B=l/tm \ T_m \ , r(\&) \ . \ : T. \ , \ : T \ p(-1+BT+ pn) \ 31 - Pm) \ (42) \] \ @?=;0$
 $1 \ 0 \ 0 \ Pm \ (T_m < T). \sim$ so-that a second-order approximation frequently used is - . E 1 7
 $t =_ T \ ?/0 \ 1 \ TI-x) \ 0 \ 0 \ Pm \ (\) \ [2]: \ Q= = 921 \ 922-923 \ (4.3) \ " \ 14931 \ 932 \ 933\} \] \ \sim 2pT$
 $2p? \ T? \ 272. \ -BT \ 7 = 5531 \ -? \ +2BT+ -2B?T? -4BTe \ qu = agile \ PT + 1 -2?787 \ 4+ \ 26 \ Te \ P?$
 $_ \ 2BT + pT] \ 113 \ >= \ 7p! \ - \ e \ PT \ _ \ 9B \ TeF?)$

202 Design and Analysis of Modern Tracking Systems – 922 = aps te P* – 3 – 6 7PT + 287]
1 __ gs = saz le PT F120 PT) 1 = 1 fy, – 2AT 433 ap! er' | . : (T ≪ T_m), (4.3) ;
202 lim Q=–* pI-0 Tm 3 ai oft sh Si of ool Ny Nf ofa , (4.2) : 0 | (t_m < T), ,
. BT>0 (ee ee) iT rm 7 – – 2 3 2 =|) I]. Q= 202m BoP (44) ; 2 [3, 4] , , .
τ_a > 10^{–7}, . , , () , (4.4).

Modeling and Tracking Dynamic Targets | 203 ; 4.6. 4.2.2 , : $a(t)$
 $= w(t)$ (4.5) $w(t) : E[w(t)] = 0, E[w(t)w(t)] = \sigma^2 \delta(t - \tau)$ [1], $a = [F \ aC]$ (4.6a) os ,
 (4.6a) , T (T_m). q $q = 202m$ q , , 7 [3]. , $q = 1002$.
 , . , , 13.

204 - Design and Analysis of Modern Tracking Systems (CA), (white jerk) , $a,(t)$
 $= w(t)$, $T_m \gg T$: $T > 74$ TF3 20 8 6 . $Q = |F| \text{ LF Lig (4.6b) } 73$ T? oc a ff , ,
 [1]. Δt Q , : $= (74 \ 7)$, 47 , [1], a , @,, 1S 05am < Oy XS am $T4/4$ 73/2
 $T?/2$ $Q = |7/2$ T? T |o; (4.7b) $T?/2 = T$ 1 [1] $A_{a,\max}$ T , $0.5Aa,, < dy < Aap$, (4.7a) ,
 , (4.7b) .

Modeling and Tracking Dynamic Targets 205 , (4.6a) (4.6b), (4.7a) (4.7b), , $(\Delta t > 0)$,
 . , Q (4.6a), [7 (3.25e) , $Q(2T) = Q(T) + Q(T)$ o! , $2T$, T , $2T$. [q
 in (4.6) a in (4.7)] , . , [3] 13. , - [1] (4.7). - 4.2.3 , , ,
 x, y, z . , , . , . Maybeck et al. [5] 13 Bar-Shalom [6], a w , v
 w : $a = -w'v + w$ (4.8) w , (4.8), : $o = v \times a_l$ (49) y^2

206 Design and Analysis of Modern Tracking Systems (4.8) (4.9) , , , . 3D-
 $s = 4$: $V_y = \cosh$, $v_y = \sinh$ (4.10) , . s , (4.10), $\therefore a_y = W_y$, $a_y = -s \cosh = -w v$, $dy =$
 $-s w' \cosh = -w' v$, $- \frac{5}{2} \frac{2}{2} a_y = W_y$, $a_y = -W^* d$, $w = / =$. . @ = 4 : $X x(k+1)$
 $= x(k) + f(v) \frac{dt}{te} \sim . : x(k) = x(k) + sT[SW \cosh - CW \sin 4] = x(k) + T[SW v, (k) - CW v, (&)]$
 $\sim$ (4.11) $-y(R+1) = y(2) + s5T[SW \sinh + CW \cos h] = y(k) + T[CW v, (&) + SW v, (2)]$ (4.12) $\sim$
 $\sin wT \frac{1}{1} - T \sin y \sin cw A \cos @$ (4.13) $\circ T wT v(k+1) = 5 \cos(h + WT) = v(A) \cos wT -$
 $v, (4) \sin wT v_y(k+1) = \sin(a + \circ T) = v, (k) \sin wT + v, (k) \cos wT$ (4.14)

Modeling and Tracking Dynamic Targets ? an 207 (4.11)(4.14) , . , () w ,
 4.3, , $(x, v_x, y, v_y, \dot{\psi})$, , (z) . , , , , [6, 9, 10] , , (4.8) w ,
 (4.9). , w . 3D- , [9] . , , . , , [11] , $(a - v \approx 0)$, 3. ,
 [12] , . [13] , (4.1) (4.8). [13]. 4.2.4 - . ,
 , , , . [14] , [15] , . , [11], , - . .